Lossy files are smaller than lossless files which is of great benefit when considering storage and data transfer rate requirements.

Common lossy file compression algorithms are:

- MPEG-3 (MP3) and MPEG-4 (MP4)
- JPEG.

MPEG-3 (MP3) and MPEG-4 (MP4)

MP3 files are used for playing music on computers or mobile phones. This compression technology will reduce the size of a normal music file by about 90%. While MP3 music files can never match the sound quality found on a DVD or CD, the quality is satisfactory for most general purposes.

But how can the original music file be reduced by 90% while still retaining most of the music quality? Essentially the algorithm removes sounds that the human ear can't hear properly. For example:

- removal of sounds outside the human ear range
- if two sounds are played at the same time, only the louder one can be heard by the ear, so the softer sound is eliminated. This is called perceptual music shaping.

MP4 files are slightly different to MP3 files. This format allows the storage of multimedia files rather than just sound – music, videos, photos and animation can all be stored in the MP4 format. As with MP3, this is a lossy file compression format, but it still retains an acceptable quality of sound and video. Movies, for example, could be streamed over the internet using the MP4 format without losing any real discernible quality.

JPEG

When a camera takes a photograph, it produces a raw bitmap file which can be very large in size. These files are temporary in nature. **JPEG** is a lossy file compression algorithm used for bitmap images. As with MP3, once the image is subjected to the JPEG compression algorithm, a new file is formed and the original file can no longer be constructed.

The JPEG file reduction process is based on two key concepts:

- human eyes don't detect differences in colour shades quite as well as they detect differences in image brightness (the eye is less sensitive to colour variations than it is to variations in brightness)
- by separating pixel colour from brightness, images can be split into 8 × 8 pixel blocks, for example, which then allows certain 'information' to be discarded from the image without causing any real noticeable deterioration in quality.

Lossless file compression

With this technique, all the data from the original uncompressed file can be reconstructed. This is particularly important for files where any loss of data would be disastrous (e.g. when transferring a large and complex spreadsheet or when downloading a large computer application).